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VeritasChain Protocol (VCP) Specification

Version 1.1

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Maintainer: VeritasChain Standards Organization (VSO)

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Website: <https://veritaschain.org>

Revision History

Version	Date	Changes	Author
1.1	2025-12-30	Three-layer architecture, External Anchor mandatory, Policy Identification, VCP-XREF, Completeness Guarantees	VSO Technical Committee
1.0	2025-11-25	Initial release	VSO Technical Committee

Summary of Changes from v1.0

Breaking Changes

Certification-level breaking changes only.

VCP v1.1 introduces new mandatory certification requirements (External Anchor and Policy Identification) that affect VC-Certified status.

Change	Protocol Compatibility	Certification Impact
PrevHash → OPTIONAL	✅ Fully compatible	No impact (relaxation)
External Anchor → REQUIRED	✅ Fully compatible	⚠️ Silver tier must add anchoring
Policy Identification → REQUIRED	✅ Fully compatible	⚠️ All tiers must add field

Summary: Existing v1.0 implementations remain **protocol-compatible** (can interoperate with v1.1 systems), but may require additional components to obtain **v1.1 VC-Certified status**.

※ v1.1 is a **protocol-compatible / certification-stricter** update.

Major Changes

#	Change	Impact	Migration
1	Three-Layer Architecture	Section 6 restructured	Documentation only
2	PrevHash now OPTIONAL	Hash chain linking no longer required	None (relaxation)
3	External Anchor REQUIRED for all tiers	Silver tier must implement daily anchoring	Implementation required
4	Policy Identification added	New Section 5.5	Implementation required
5	VCP-XREF Dual Logging added	New Section 5.6	OPTIONAL extension

Rationale

The v1.0 specification prioritized flexibility for Silver tier implementations by making External Anchor optional. However, community feedback identified that without mandatory external anchoring, the “Verify, Don’t Trust” principle could not be fully realized—log producers could theoretically modify Merkle Roots before anchoring.

v1.1 strengthens this by: 1. Making External Anchor REQUIRED for all tiers (with tier-appropriate frequencies and lightweight options for Silver) 2. Clarifying that hash chains (PrevHash) are an OPTIONAL

local integrity mechanism that complements, but does not replace, external verifiability 3. Establishing a clear three-layer architecture that separates concerns and clarifies where integrity guarantees originate

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1. Introduction

1.1 Purpose

The VeritasChain Protocol (VCP) is a global standard specification for recording the “decision-making” and “execution results” of algorithmic trading in an immutable and verifiable format. VCP provides a cryptographically secured chain of evidence that establishes truth (“Veritas”) in trading operations, ensuring compliance with international regulations including MiFID II, GDPR, EU AI Act, and emerging quantum-resistant security requirements.

Completeness Guarantees (NEW in v1.1): VCP v1.1 extends tamper-evidence to **completeness guarantees**, enabling third parties to cryptographically verify not only that events were not altered, but that **no required events were omitted** (omission / split-view attacks). This is achieved through mandatory Merkle Tree construction and external anchoring for all tiers, ensuring that event batches are provably complete at the time of anchoring.

1.2 Scope

VCP applies to: - **High-Frequency Trading (HFT) systems** - **Algorithmic and AI-driven trading platforms** - **Retail trading systems (MT4/MT5)** - **Cryptocurrency exchanges** - **Regulatory reporting systems**

1.3 Versioning

VCP adopts Semantic Versioning 2.0.0: - **MAJOR** version: Incompatible API changes - **MINOR** version: Backwards-compatible functionality additions - **PATCH** version: Backwards-compatible bug fixes

Full backward compatibility is guaranteed within the v1.x series.

1.4 Crypto Agility

VCP implements crypto agility to ensure future-proof security: - **Current Default:** Ed25519 (performance and security optimized) - **Supported Algorithms:** Ed25519, ECDSA_SECP256K1, RSA_2048 - **Future Reserved:** Post-quantum algorithms (DILITHIUM, FALCON) - **Migration Path:** Automated algorithm upgrade capability

1.5 Standard Enumerations

1.5.1 SignAlgo Enum

Value	Algorithm	Description	Status
ED25519	Ed25519	Edwards-curve Digital Signature	DEFAULT
ECDSA_SECP256K1	ECDSA secp256k1	Bitcoin/Ethereum compatible	SUPPORTED
RSA_2048	RSA 2048-bit	Legacy systems	DEPRECATED
DILITHIUM2	CRYSTALS-Dilithium	Post-quantum (NIST Level 2)	FUTURE
FALCON512	FALCON-512	Post-quantum (NIST Level 1)	FUTURE

1.5.2 HashAlgo Enum

Value	Algorithm	Description	Status
SHA256	SHA-256	SHA-2 family, 256-bit	DEFAULT
SHA3_256	SHA3-256	SHA-3 family, 256-bit	SUPPORTED
BLAKE3	BLAKE3	High-performance hash	SUPPORTED
SHA3_512	SHA3-512	SHA-3 family, 512-bit	FUTURE

1.5.3 ClockSyncStatus Enum

Value	Description	Tier Applicability
PTP_LOCKED	PTP synchronized with lock	Platinum

Value	Description	Tier Applicability
NTP_SYNCED	NTP synchronized	Gold
BEST_EFFORT	Best-effort synchronization	Silver
UNRELIABLE	No reliable synchronization	Silver (degraded)

1.5.4 TimestampPrecision Enum

Value	Description	Decimal Places
NANOSECOND	Nanosecond precision	9
MICROSECOND	Microsecond precision	6
MILLISECOND	Millisecond precision	3

1.6 Core Modules

- **VCP-CORE**: Standard header and security layer
- **VCP-TRADE**: Trading data payload schema
- **VCP-GOV**: Algorithm governance and AI transparency
- **VCP-RISK**: Risk management parameter recording
- **VCP-PRIVACY**: Privacy protection with crypto-shredding
- **VCP-RECOVERY**: Chain disruption recovery mechanism
- **VCP-XREF**: Cross-reference and dual logging (NEW in v1.1)

2. Compliance Tiers

2.1 Tier Definitions

Tier	Target	Clock Sync	Serialization	Signature	External Anchor	Precision
Platinum	HFT/ Exchange	PTPv2 ($<1\mu\text{s}$)	SBE	Ed25519 (Hardware)	REQUIRED (10 min)	NANOSECOND
Gold	Prop/ Institutional	NTP ($<1\text{ms}$)	JSON	Ed25519 (Client)	REQUIRED (1 hour)	MICROSECOND
Silver	Retail/ MT4/5	Best- effort	JSON	Ed25519 (Delegated)	REQUIRED (24 hours)	MILLISECOND

CHANGE FROM v1.0: External Anchor is now REQUIRED for all tiers to ensure externally verifiable integrity. For Silver tier, lightweight mechanisms (e.g., OpenTimestamps, FreeTSA) are explicitly acceptable. This change aligns all tiers with VCP's "Verify, Don't Trust" principle.

2.2 Tier-Specific Requirements

2.2.1 Platinum Tier

```
Requirements:
Clock:
  Protocol: PTPv2 (IEEE 1588-2019)
  Accuracy: <1 microsecond
  Status: PTP_LOCKED required
Performance:
  Throughput: >1M events/second
  Latency: <10µs per event
  Storage: Binary (SBE/FlatBuffers)
ExternalAnchor:
  Frequency: Every 10 minutes
  Target: Blockchain or RFC 3161 TSA
  ProofType: Full Merkle proof
Implementation:
  Languages: [C++, Rust, FPGA]
  Techniques: [Kernel bypass, RDMA, Zero-copy]
```

2.2.2 Gold Tier

```
Requirements:
Clock:
  Protocol: NTP/Chrony
  Accuracy: <1 millisecond
  Status: NTP_SYNCED required
Performance:
  Throughput: >100K events/second
  Latency: <100µs per event
  Persistence: WAL/Queue required (Kafka, Redis)
ExternalAnchor:
  Frequency: Every 1 hour
  Target: RFC 3161 TSA or Database with third-party attestation
  ProofType: Merkle root + audit path
Implementation:
  Languages: [Python, Java, C#]
  Deployment: Cloud-ready (AWS/GCP/Azure)
```

2.2.3 Silver Tier

```
Requirements:
Clock:
```

Protocol: System time
Accuracy: Best-effort
Status: BEST_EFFORT/UNRELIABLE accepted
Performance:
Throughput: >1K events/second
Latency: <1 second
Communication: Async recommended
ExternalAnchor:
Frequency: Every 24 hours (daily)
Target: Database with integrity proof or public timestamping service
ProofType: Merkle root only
Implementation:
Languages: [MQL5, Python]
Compatibility: MT4/MT5 DLL integration

NOTE: Silver tier is NOT intended for regulatory-grade algorithmic trading systems subject to MiFID II RTS 25, SEC Rule 17a-4, or equivalent clock synchronization requirements. Silver tier is appropriate for development, testing, backtesting analysis, and non-regulated trading scenarios.

GUIDANCE on Silver Tier for Semi-Regulatory Use Cases: In practice, Silver tier logs may be used in contexts with indirect regulatory implications (e.g., presenting backtesting results to supervisory authorities, internal audit documentation). In such cases:

Aspect	Silver Tier Capability	Reasonable Assurance Level
Timestamp accuracy	BEST_EFFORT (system clock)	Indicative only; not suitable for latency disputes
Event completeness	Daily Merkle anchor	Batch-level integrity; gaps possible within 24h window
Chain continuity	PrevHash OPTIONAL	No real-time gap detection unless enabled

For higher assurance within the 24-hour window, implementations MAY perform **intraday manual anchoring** (e.g., at end of trading session) or reduce the anchor interval to 12 hours. This does not change the tier classification but improves auditability.

Completeness Guarantee Scope: Silver tier provides batch-level completeness guarantees at anchor time, not continuous real-time completeness.

Organizations using Silver tier logs for regulatory explanations should clearly disclose these limitations to authorities. For higher assurance, upgrade to Gold tier.

3. Event Lifecycle

[Section 3 unchanged from v1.0]

3.1 Event State Diagram

```
stateDiagram-v2
    [*] --> INIT: System Start
    INIT --> SIG: Signal Generated

    SIG --> ORD: Order Sent
    SIG --> REJ: Rejected (Risk)

    ORD --> ACK: Acknowledged
    ORD --> REJ: Rejected (Broker)

    ACK --> EXE: Filled (Full)
    ACK --> PRT: Partial Fill
    ACK --> CXL: Cancelled

    PRT --> EXE: Remaining Filled
    PRT --> CXL: Remaining Cancelled

    EXE --> MOD: Modified
    EXE --> CLS: Position Closed
```

3.2 Event Type Registry

[Event Type Registry unchanged from v1.0 - see v1.0 specification for complete list]

4. Data Model

[Section 4 unchanged from v1.0]

5. Extension Modules

[Sections 5.1-5.4 unchanged from v1.0]

5.5 Policy Identification (NEW in v1.1)

5.5.1 Purpose

Policy Identification ensures that every VCP event explicitly declares its conformance tier and registration policy. This enables verifiers to apply appropriate validation rules and supports multi-tier deployments.

5.5.2 Schema Definition

```
{
  "PolicyIdentification": {
    "Version": "1.1",
    "PolicyID": "string",           // REQUIRED: Unique policy identifier
    "ConformanceTier": "enum",     // REQUIRED: SILVER | GOLD | PLATINUM
    "RegistrationPolicy": {
      "Issuer": "string",          // Organization operating the policy
      "PolicyURI": "string",       // URI to policy document
      "EffectiveDate": "int64",    // Policy effective timestamp
      "ExpirationDate": "int64"   // Policy expiration (optional)
    },
    "VerificationDepth": {
      "HashChainValidation": "boolean", // Whether hash chain is used
      "MerkleProofRequired": "boolean", // Always true in v1.1
      "ExternalAnchorRequired": "boolean" // Always true in v1.1
    }
  }
}
```

5.5.3 Requirements

Field	Requirement	Description
PolicyID	REQUIRED	Unique identifier for the registration policy
ConformanceTier	REQUIRED	Must be one of: SILVER, GOLD, PLATINUM
RegistrationPolicy.Issuer	REQUIRED	Organization name or identifier
VerificationDepth	REQUIRED	Declares verification capabilities

5.5.4 PolicyID Naming Convention (NEW in v1.1)

To ensure global uniqueness without requiring a central registry, PolicyID SHOULD follow the **Issuer Domain + Local ID** format:

PolicyID = <reverse_domain>:<local_identifier>

Examples:

org.veritaschain.prod:hft-system-001
com.example.trading:gold-algo-v2
jp.co.broker:silver-mt5-bridge

Component	Format	Example
Reverse Domain	Reverse DNS notation of Issuer's domain	org.veritaschain
Separator	Colon (:)	:
Local Identifier	Issuer-defined, alphanumeric with hyphens	prod-hft-001

NOTE: VSO does not operate a PolicyID registry. Uniqueness is achieved through domain ownership. Organizations without domains MAY use `local:<organization_name>:<local_id>` format, but this provides weaker uniqueness guarantees.

5.5.5 Relationship to Conformance Tiers

Conformance tiers (Silver/Gold/Platinum) represent **verification depth**, not separate registration policies. A single organization may operate multiple policies at different tiers for different use cases:

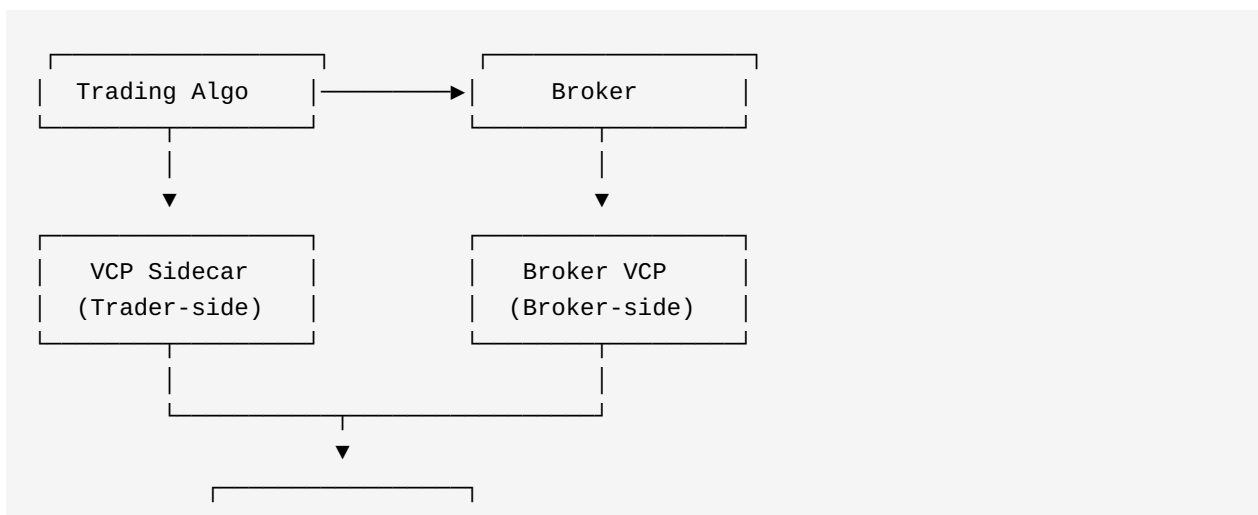
- **Platinum:** Production HFT systems
- **Gold:** Standard algorithmic trading
- **Silver:** Development, testing, backtesting

The Registration Policy identifier MUST be explicitly included in the payload or metadata of every VCP event.

5.6 VCP-XREF: Cross-Reference and Dual Logging (NEW in v1.1)

5.6.1 Purpose

VCP-XREF enables **Dual Logging**—independent VCP event streams from multiple parties that can be cross-referenced to detect discrepancies. This provides a higher level of assurance than single-party logging by ensuring that collusion between parties is required to manipulate records undetected.



Cross-Reference
Verification

Guarantee: Unless both parties collude,
omission or modification by one party
is detectable by the other.

5.6.2 Use Cases

Scenario	Party A	Party B	Benefit
Prop Firm Trading	Trader	Prop Firm	Prevent payout disputes
Broker Execution	Algo Provider	Broker	Verify best execution
Multi-Venue	Smart Order Router	Exchange	Cross-venue audit
Regulatory Audit	Trading Firm	Regulator	Independent verification

5.6.3 Schema Definition

```
{
  "VCP-XREF": {
    "Version": "1.1",
    "CrossReferenceID": "uuid",           // REQUIRED: Shared reference ID
    "PartyRole": "enum",                 // REQUIRED: INITIATOR | COUNTERPARTY |
OBSERVER
    "CounterpartyID": "string",           // REQUIRED: Identifier of the other
party
    "SharedEventKey": {
      "OrderID": "string",               // Primary correlation key
      "AlternateKeys": ["string"],        // Additional correlation keys
      "Timestamp": "int64",               // Event timestamp for matching
      "ToleranceMs": "int32"              // Timestamp matching tolerance
    },
    "ExpectedCounterpartyHash": "string", // OPTIONAL: Expected hash from
counterparty
    "ReconciliationStatus": "enum",        // PENDING | MATCHED | DISCREPANCY |
TIMEOUT
    "DiscrepancyDetails": {
      "Field": "string",
      "LocalValue": "string",
      "CounterpartyValue": "string",
      "Severity": "enum"                  // INFO | WARNING | CRITICAL
    }
  }
}
```

```
}  
}
```

5.6.4 Party Roles

Role	Description	Responsibilities
INITIATOR	Party that initiates the transaction	Generate CrossReferenceID, log first
COUNTERPARTY	Party that receives/executes the transaction	Reference CrossReferenceID, log response
OBSERVER	Third-party observer (e.g., regulator)	Read-only cross-reference access

5.6.5 Cross-Reference Protocol

Step 1: Initiator Logs Event

```
{  
  "Header": {  
    "EventID": "019abc...",  
    "EventType": "ORD"  
  },  
  "VCP-XREF": {  
    "CrossReferenceID": "550e8400-e29b-41d4-a716-446655440000",  
    "PartyRole": "INITIATOR",  
    "CounterpartyID": "broker.example.com",  
    "SharedEventKey": {  
      "OrderID": "ORD-2025-001234",  
      "Timestamp": 1735084800123456789,  
      "ToleranceMs": 100  
    },  
    "ReconciliationStatus": "PENDING"  
  }  
}
```

Step 2: Counterparty Logs Event

```
{  
  "Header": {  
    "EventID": "019def...",  
    "EventType": "ACK"  
  },  
  "VCP-XREF": {  
    "CrossReferenceID": "550e8400-e29b-41d4-a716-446655440000",  
    "PartyRole": "COUNTERPARTY",  
    "CounterpartyID": "trader.example.com",  
  }  
}
```

```

        "SharedEventKey": {
            "OrderID": "ORD-2025-001234",
            "Timestamp": 1735084800123789012,
            "ToleranceMs": 100
        },
        "ExpectedCounterpartyHash": "sha256:abc123...",
        "ReconciliationStatus": "MATCHED"
    }
}

```

Step 3: Cross-Reference Verification

Either party (or a third-party auditor) can verify:

```

def verify_cross_reference(initiator_event, counterparty_event):
    """
    Verify that both parties logged consistent events
    """
    # Check CrossReferenceID matches
    if initiator_event["VCP-XREF"]["CrossReferenceID"] != \
        counterparty_event["VCP-XREF"]["CrossReferenceID"]:
        return "DISCREPANCY: CrossReferenceID mismatch"

    # Check SharedEventKey matches within tolerance
    time_diff = abs(
        initiator_event["VCP-XREF"]["SharedEventKey"]["Timestamp"] -
        counterparty_event["VCP-XREF"]["SharedEventKey"]["Timestamp"]
    )
    tolerance = initiator_event["VCP-XREF"]["SharedEventKey"]["ToleranceMs"] *
1_000_000

    if time_diff > tolerance:
        return "DISCREPANCY: Timestamp outside tolerance"

    # Check order details match
    if initiator_event["VCP-XREF"]["SharedEventKey"]["OrderID"] != \
        counterparty_event["VCP-XREF"]["SharedEventKey"]["OrderID"]:
        return "DISCREPANCY: OrderID mismatch"

    return "MATCHED"

```

5.6.6 Discrepancy Handling

When cross-reference verification detects a discrepancy:

Severity	Example	Action
INFO	Timestamp diff within 2x tolerance	Log for monitoring

Severity	Example	Action
WARNING	Minor field difference (e.g., rounding)	Alert operations
CRITICAL	Order existence dispute, price mismatch	Escalate to dispute resolution

5.6.7 Security Considerations

Threat	Mitigation
Single-party manipulation	Counterparty log provides independent evidence
Collusion	External anchoring makes post-hoc collusion detectable
Replay attacks	CrossReferenceID + Timestamp uniqueness
Denial of logging	Missing counterparty record is itself evidence

KEY GUARANTEE: If Party A claims an event occurred and Party B denies it, the presence or absence of VCP-XREF records from both parties provides **non-repudiable evidence**. Manipulation requires collusion between both parties AND compromise of external anchors.

5.6.8 Requirements

Requirement	Level	Notes
VCP-XREF Extension	OPTIONAL	RECOMMENDED for dispute-prone scenarios
CrossReferenceID format	UUID v4 or v7	MUST be globally unique
SharedEventKey	At least one key	OrderID recommended as primary
External Anchor	REQUIRED	Both parties must anchor independently
Retention	Match regulatory minimum	Typically 7 years

5.6.9 Relationship to External Anchor

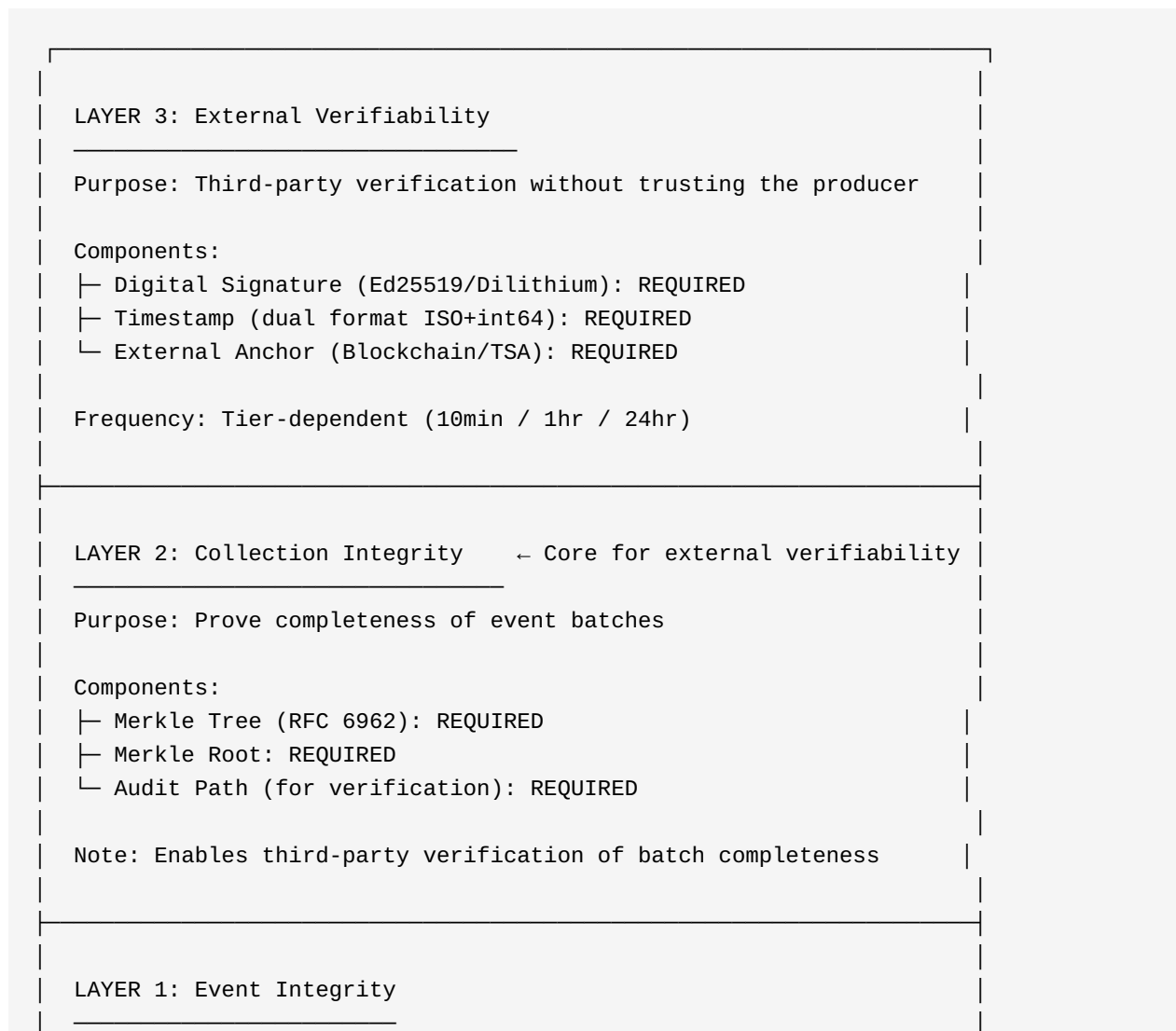
VCP-XREF is **complementary** to External Anchoring, not a replacement:

Mechanism	Provides	Limitation
External Anchor	Tamper evidence for single party	Single party could omit events before anchoring
VCP-XREF	Cross-party verification	Requires counterparty cooperation
Both combined	Maximum assurance	Collusion + anchor compromise required to manipulate

6. Integrity and Security Layer (Three-Layer Architecture)

6.0 Architectural Overview (NEW in v1.1)

VCP v1.1 introduces a clear three-layer architecture for integrity and security. This structure clarifies the relationship between different cryptographic mechanisms and their roles in ensuring audit trail integrity.



Purpose: Individual event completeness
Components:
└ EventHash (SHA-256 of canonical event): REQUIRED
└ PrevHash (link to previous event): OPTIONAL
Note: PrevHash provides real-time detection (OPTIONAL in v1.1)

6.0.1 Layer Responsibilities

Layer	Purpose	REQUIRED Components	OPTIONAL Components
Layer 3	External Verifiability	Signature, Timestamp, External Anchor	Dual signatures (PQC)
Layer 2	Collection Integrity	Merkle Tree, Merkle Root, Audit Path	-
Layer 1	Event Integrity	EventHash	PrevHash (hash chain)

6.0.2 Why This Architecture?

Question: Why is hash chain (PrevHash) OPTIONAL in v1.1 while it was REQUIRED in v1.0?

Answer:

PrevHash-based hash chaining was REQUIRED in v1.0 to prioritize real-time, in-process tamper detection. This remains a valid and valuable integrity mechanism.

In v1.1, PrevHash is OPTIONAL because equivalent or stronger integrity guarantees can be achieved through Merkle-based collection integrity (Layer 2) combined with mandatory external anchoring (Layer 3).

This change: 1. **Simplifies Silver tier implementations** without sacrificing external verifiability 2. **Aligns with the “Verify, Don’t Trust” principle** by emphasizing externally verifiable proofs 3. **Maintains full backward compatibility** with v1.0 implementations that use hash chains

Implementations that benefit from real-time tamper detection (e.g., HFT systems) SHOULD continue to use PrevHash.

6.1 Layer 1: Event Integrity

6.1.1 EventHash Calculation (REQUIRED)

Every VCP event MUST include an EventHash computed over its canonical form.

```

def calculate_event_hash(header: dict, payload: dict, algo: str = "SHA256") -> str:
    """
    Calculate event hash with RFC 8785 canonicalization

    REQUIRED for all VCP events
    """
    # Step 1: Canonicalize JSON (RFC 8785 JCS)
    canonical_header = canonicalize_json(header)
    canonical_payload = canonicalize_json(payload)

    # Step 2: Concatenate components
    hash_input = canonical_header + canonical_payload

    # Step 3: Apply hash function
    if algo == "SHA256":
        return hashlib.sha256(hash_input.encode()).hexdigest()
    elif algo == "SHA3_256":
        return hashlib.sha3_256(hash_input.encode()).hexdigest()
    elif algo == "BLAKE3":
        return blake3(hash_input.encode()).hexdigest()
    else:
        raise ValueError(f"Unsupported hash algorithm: {algo}")

```

6.1.2 Hash Chain Linking (OPTIONAL)

Implementations MAY include PrevHash to create a hash chain for real-time tamper detection.

```

def calculate_event_hash_with_chain(
    header: dict,
    payload: dict,
    prev_hash: str, # OPTIONAL: ""*64 if not using chain
    algo: str = "SHA256"
) -> str:
    """
    Calculate event hash with optional chain linking

    OPTIONAL: Use when real-time tamper detection is desired
    """
    canonical_header = canonicalize_json(header)
    canonical_payload = canonicalize_json(payload)

    # Include prev_hash in calculation if provided
    if prev_hash and prev_hash != "" * 64:
        hash_input = canonical_header + canonical_payload + prev_hash
    else:
        hash_input = canonical_header + canonical_payload

    return hash_function(hash_input, algo)

```

6.1.3 When to Use Hash Chains

Use Case	Hash Chain Recommended?	Rationale
HFT Systems	Yes	Real-time detection of event loss
Regulatory Submission	Yes	Familiar to auditors
Development/Testing	No	Simplifies implementation
Backtesting Analysis	No	Events may be generated out of order
MT4/MT5 Integration	No	Reduces DLL complexity

6.2 Layer 2: Collection Integrity

6.2.1 Merkle Tree Construction (REQUIRED)

All VCP implementations MUST construct Merkle Trees over event batches.

MANDATORY: Merkle tree construction MUST follow RFC 6962 to prevent second preimage attacks:

```
def merkle_hash(data: bytes, leaf: bool = True) -> bytes:
    """
    RFC 6962 compliant Merkle tree hashing

    REQUIRED for all VCP implementations
    """
    if leaf:
        # Leaf nodes: 0x00 prefix
        return hashlib.sha256(b'\x00' + data).digest()
    else:
        # Internal nodes: 0x01 prefix
        return hashlib.sha256(b'\x01' + data).digest()

def build_merkle_tree(event_hashes: List[str]) -> MerkleTree:
    """
    Build RFC 6962 compliant Merkle tree from event hashes

    REQUIRED for all VCP implementations
    """
    # Convert hex strings to bytes
    leaves = [bytes.fromhex(h) for h in event_hashes]

    # Build tree with domain separation
    tree = []
    current_level = [merkle_hash(leaf, leaf=True) for leaf in leaves]
    tree.append(current_level)
```

```

while len(current_level) > 1:
    next_level = []
    for i in range(0, len(current_level), 2):
        if i + 1 < len(current_level):
            combined = current_level[i] + current_level[i + 1]
        else:
            combined = current_level[i] + current_level[i] # Duplicate odd

            next_level.append(merkle_hash(combined, leaf=False))
    tree.append(next_level)
    current_level = next_level

return MerkleTree(root=current_level[0], levels=tree)

```

node

6.2.2 Merkle Root (REQUIRED)

Every batch of VCP events MUST produce a Merkle Root that is externally anchored according to the tier requirements.

```

def get_merkle_root(tree: MerkleTree) -> str:
    """
    Get the Merkle root as hex string

    REQUIRED for external anchoring
    """
    return tree.root.hex()

```

6.2.3 Audit Path Generation (REQUIRED)

Implementations MUST be able to generate audit paths for individual event verification.

```

def generate_audit_path(tree: MerkleTree, leaf_index: int) -> List[str]:
    """
    Generate RFC 6962 audit path for a specific event

    REQUIRED for verification
    """
    path = []
    index = leaf_index

    for level in tree.levels[::-1]: # Exclude root level
        sibling_index = index ^ 1 # XOR to get sibling
        if sibling_index < len(level):
            path.append({
                "hash": level[sibling_index].hex(),
                "position": "left" if sibling_index < index else "right"
            })
        index //= 2

```

```
return path
```

6.3 Layer 3: External Verifiability

6.3.1 Digital Signatures (REQUIRED)

All VCP events MUST be digitally signed.

SignAlgo Enum	Use Case	Key Size	Performance	Quantum-Resistant
ED25519	Default	256-bit	Fastest	No
ECDSA_SECP256K1	Bitcoin compatibility	256-bit	Fast	No
RSA_2048	Legacy systems	2048-bit	Slow	No
DILITHIUM2	Future (reserved)	2420 bytes	Medium	Yes
FALCON512	Future (reserved)	897 bytes	Fast	Yes

NOTE on Quantum Resistance: Post-quantum mechanisms are OPTIONAL and intended for experimental or high-assurance deployments. DILITHIUM2 and FALCON512 are reserved for future use pending wider ecosystem support.

```
def sign_merkle_root(merkle_root: str, private_key: bytes, algo: str = "ED25519") -
> str:
    """
    Sign the Merkle root for external anchoring

    REQUIRED for all VCP implementations
    """
    if algo == "ED25519":
        signing_key = Ed25519SigningKey(private_key)
        signature = signing_key.sign(bytes.fromhex(merkle_root))
        return base64.b64encode(signature).decode()
    else:
        raise ValueError(f"Unsupported signature algorithm: {algo}")
```

6.3.2 Timestamps (REQUIRED)

VCP requires dual-format timestamps for maximum compatibility:

```
{  
  "TimestampISO": "2025-12-25T12:00:00.123456789Z",  
  "TimestampInt": 1735128000123456789  
}
```

Precision vs. Accuracy: This specification distinguishes timestamp resolution from clock accuracy. Nanosecond-resolution timestamps represent the storage format capability, while actual clock accuracy is explicitly recorded in the ClockSyncStatus field and enforced per tier.

6.3.3 External Anchoring (REQUIRED - Changed in v1.1)

CRITICAL CHANGE: External Anchoring is REQUIRED for all tiers to achieve externally verifiable integrity.

For Silver tier, lightweight or delegated anchoring mechanisms are explicitly acceptable and expected. The requirement ensures that even the simplest VCP implementation provides third-party verifiable proof of integrity.

Tier	Frequency	Anchor Target	Proof Type
Platinum	10 minutes	Blockchain (Ethereum, etc.) or RFC 3161 TSA	Full Merkle proof
Gold	1 hour	RFC 3161 TSA or attested database	Merkle root + audit path
Silver	24 hours	Public timestamping service or attested database	Merkle root only

Acceptable Anchor Targets

Target Type	Description	Tier Applicability
Public Blockchain	Ethereum, Bitcoin, etc.	Platinum
RFC 3161 TSA	Qualified Time Stamp Authority	Platinum, Gold
Attested Database	Database with third-party attestation	Gold, Silver
Public Timestamping Service	Free/public services (e.g., OpenTimestamps)	Silver

Attested Database Requirements (NEW in v1.1)

For “Attested Database” to qualify as an anchor target, it MUST meet the following minimum criteria:

Criterion	Requirement	Verification
Third-Party Audit	Annual audit by independent party	Audit report available
Tamper Detection	Cryptographic integrity checks (hash chain, Merkle, or equivalent)	Technical documentation
Access Controls	Role-based access with audit logging	SOC 2 Type II or equivalent
Retention Policy	Data retention $\geq$ regulatory minimum (typically 7 years)	Policy documentation
Availability SLA	$\geq 99.9\%$ uptime commitment	SLA documentation

NOTE: “Public Timestamping Service” (Silver tier) has no attestation requirement but provides weaker assurance. For regulatory use cases, Attested Database or higher is recommended.

Attested Database Examples (Non-Exhaustive)

Example	Attestation Level	Notes
AWS QLDB + SOC 2 Type II	High	Immutable ledger with annual audit
Azure SQL Ledger + SOC 2	High	Cryptographic verification built-in
Google Cloud Spanner + SOC 2	High	Distributed database with audit trail
Self-hosted PostgreSQL + annual crypto audit	Medium	Requires third-party hash verification
Internal database without attestation	Not acceptable	Does not meet “Attested” criteria

For detailed attestation acceptance criteria, see: **VSO-CAB-REQ-001** (CAB Accreditation Requirements)

Anchor Target Unavailability (NEW in v1.1)

Implementations **MUST** handle anchor target unavailability:

Scenario	Required Action
Temporary outage	Queue anchoring requests; retry with exponential backoff
Permanent discontinuation	Migrate to alternative anchor target within 30 days
Anchor verification failure	Retain local AnchorRecord copy as backup proof

Implementations SHOULD maintain a local complete copy of all AnchorRecords to enable verification even if the original anchor target becomes unavailable.

Anchoring Record Schema

```

{
  "AnchorRecord": {
    "MerkleRoot": "string",           // REQUIRED: Hex-encoded root
    "Signature": "string",           // REQUIRED: Base64-encoded signature
    "SignAlgo": "ED25519",           // REQUIRED: Signature algorithm
    "Timestamp": "int64",            // REQUIRED: Anchor time
    "AnchorTarget": {
      "Type": "enum",                // BLOCKCHAIN | TSA | DATABASE |
      "Identifier": "string",         // Chain ID, TSA URL, etc.
      "Proof": "string"              // Transaction hash, TSA token, etc.
    },
    "EventCount": "int32",            // Number of events in batch
    "FirstEventID": "uuid",           // First event in batch
    "LastEventID": "uuid",            // Last event in batch
    "PolicyID": "string"              // Reference to Policy Identification
  }
}

```

Silver Tier Anchoring Options

For Silver tier implementations with limited infrastructure, the following simplified anchoring options are acceptable:

1. **OpenTimestamps**: Free, Bitcoin-backed timestamping
2. **FreeTSA**: Free RFC 3161 compliant service
3. **OriginStamp**: Commercial service with free tier
4. **Self-hosted with attestation**: Database with periodic third-party audit

```

def anchor_silver_tier(merkle_root: str, signature: str) -> AnchorRecord:
    """
    Example: Silver tier anchoring with OpenTimestamps
    """
    import opentimestamps

    # Create timestamp
    timestamp = opentimestamps.create_timestamp(bytes.fromhex(merkle_root))

```



```

return AnchorRecord(
    merkle_root=merkle_root,
    signature=signature,
    anchor_target={
        "type": "PUBLIC_SERVICE",
        "identifier": "opentimestamps.org",
        "proof": timestamp.serialize().hex()
    }
)

```

6.4 Security Object Schema (Updated in v1.1)

```

{
  "Security": {
    "Version": "1.1",
    "EventHash": "string",           // REQUIRED: SHA-256 of canonical event
    "PrevHash": "string",           // OPTIONAL: Hash of previous event (v1.1
change)
    "HashAlgo": "SHA256",           // REQUIRED: Hash algorithm used
    "Signature": "string",           // REQUIRED: Base64-encoded signature
    "SignAlgo": "ED25519",           // REQUIRED: Signature algorithm
    "MerkleRoot": "string",           // REQUIRED: Current batch Merkle root
    "MerkleIndex": "int32",           // REQUIRED: Position in Merkle tree
    "AnchorReference": "string"       // REQUIRED: Reference to anchor record
  }
}

```

6.4.1 Field Requirements by Version

Field	v1.0	v1.1	Notes
EventHash	REQUIRED	REQUIRED	No change
PrevHash	REQUIRED (except INIT)	OPTIONAL	Relaxed in v1.1
HashAlgo	REQUIRED	REQUIRED	No change
Signature	REQUIRED	REQUIRED	No change
SignAlgo	REQUIRED	REQUIRED	No change
MerkleRoot	OPTIONAL (Gold/Platinum)	REQUIRED	Strengthened in v1.1
MerkleIndex	OPTIONAL	REQUIRED	New in v1.1

Field	v1.0	v1.1	Notes
AnchorReference	OPTIONAL	REQUIRED	Strengthened in v1.1

7. Implementation Guidelines

[Section 7 largely unchanged from v1.0, with updates for three-layer architecture]

7.1 Minimum Viable Implementation by Tier

7.1.1 Silver Tier Minimum

```
class SilverTierVCP:
    """
    Minimum viable Silver tier implementation

    REQUIRED:
    - EventHash calculation
    - Merkle Tree construction (daily)
    - Digital signature
    - External anchor (daily)

    OPTIONAL:
    - Hash chain (PrevHash)
    """

    def __init__(self, private_key: bytes, policy_id: str):
        self.private_key = private_key
        self.policy_id = policy_id
        self.pending_events = []
        self.last_anchor_time = None

    def log_event(self, header: dict, payload: dict) -> dict:
        # Calculate EventHash (REQUIRED)
        event_hash = calculate_event_hash(header, payload)

        # Create security object (no PrevHash - OPTIONAL)
        security = {
            "Version": "1.1",
            "EventHash": event_hash,
            "HashAlgo": "SHA256",
            "SignAlgo": "ED25519"
        }

        event = {
            "Header": header,
            "Payload": payload,
            "Security": security,
```

```

        "PolicyIdentification": {
            "PolicyID": self.policy_id,
            "ConformanceTier": "SILVER"
        }
    }

    # Sign event (REQUIRED)
    event["Security"]["Signature"] = sign_event(event_hash, self.private_key)

    self.pending_events.append(event)
    return event

def anchor_batch(self) -> AnchorRecord:
    """
    REQUIRED: Must be called at least daily for Silver tier
    """
    if not self.pending_events:
        return None

    # Build Merkle tree (REQUIRED)
    event_hashes = [e["Security"]["EventHash"] for e in self.pending_events]
    merkle_tree = build_merkle_tree(event_hashes)
    merkle_root = get_merkle_root(merkle_tree)

    # Sign Merkle root (REQUIRED)
    root_signature = sign_merkle_root(merkle_root, self.private_key)

    # External anchor (REQUIRED in v1.1)
    anchor = anchor_silver_tier(merkle_root, root_signature)

    # Update events with Merkle info
    for i, event in enumerate(self.pending_events):
        event["Security"]["MerkleRoot"] = merkle_root
        event["Security"]["MerkleIndex"] = i
        event["Security"]["AnchorReference"] = anchor.id

    self.pending_events = []
    self.last_anchor_time = time.time()

    return anchor

```

8. Regulatory Compliance

[Core requirements unchanged from v1.0]

8.1 ClockSyncStatus Usage in Regulatory Context (NEW in v1.1)

VCP distinguishes between timestamp **precision** (storage format) and **accuracy** (clock synchronization). Regulatory evaluation should consider both:

8.1.1 Interpreting ClockSyncStatus for Compliance

ClockSyncStatus	Regulatory Interpretation	Applicable Standards
PTP_LOCKED	Authoritative timestamp; suitable for latency disputes	MiFID II RTS 25 (gateway-level)
NTP_SYNCED	Reliable timestamp; suitable for order sequencing	MiFID II RTS 25 (general), CAT Rule 613
BEST_EFFORT	Indicative timestamp; not suitable for precise sequencing	Internal audit only
UNRELIABLE	Timestamp may be significantly inaccurate	Development/testing only

8.1.2 Example: MiFID II RTS 25 Compliance Assessment

Event Log:

Timestamp: 2025-12-25T10:30:00.123456789Z

TimestampPrecision: NANOSECOND

ClockSyncStatus: NTP_SYNCED

Assessment:

- ✓ Precision: Nanosecond (exceeds RTS 25 microsecond requirement)
- ✓ Accuracy: NTP synced (~1ms) - acceptable for general trading
- ⚠ Note: For gateway timestamps, PTP_LOCKED may be required

8.1.3 Accuracy Degradation Handling

When ClockSyncStatus degrades (e.g., NTP_SYNCED → BEST_EFFORT), implementations SHOULD:

1. Log a SNC (Clock Sync) event with the status change
 2. Continue logging events with the degraded status
 3. Alert operations team for remediation
 4. Document the degradation period for regulatory inquiry
-

9. Testing Requirements

9.1 Conformance Test Suite (Updated for v1.1)

9.1.1 Tier Requirements Matrix

Test Category	Silver	Gold	Platinum
Schema Validation	Required	Required	Required
UUID v7 Format	Required	Required	Required
Timestamp (MILLISECOND)	Required	Required	Required
Timestamp (MICROSECOND)	Optional	Required	Required
Timestamp (NANOSECOND)	Optional	Optional	Required
EventHash Calculation	Required	Required	Required
Hash Chain (PrevHash)	Optional	Optional	Optional
Digital Signature	Required	Required	Required
Merkle Tree Construction	Required	Required	Required
Merkle Proof Verification	Required	Required	Required
External Anchor	Required	Required	Required
Policy Identification	Required	Required	Required
Clock Sync (BEST_EFFORT)	Required	Required	Required
Clock Sync (NTP_SYNCED)	Optional	Required	Required
Clock Sync (PTP_LOCKED)	Optional	Optional	Required

CHANGES FROM v1.0: - Hash Chain: Changed from Required to Optional for all tiers -
External Anchor: Changed from Optional/Recommended/Required to Required for all tiers -
Policy Identification: New requirement for all tiers

NOTE on Merkle Proof Verification for Silver Tier: For Silver tier, “Merkle Proof Verification” refers to the ability to verify that a given EventHash is included in the anchored Merkle Root (batch-level verification). Per-event audit path **storage** MAY be omitted; however, implementations **MUST** retain sufficient data to **generate audit paths on-demand** when requested for audit or regulatory inquiry. Gold and Platinum tiers **MUST** support full per-event audit path generation and storage.

9.1.2 Critical Tests

Some tests are marked **CRITICAL**. Any failure in a critical test results in automatic certification failure.

Critical tests in v1.1:

Test ID	Description	v1.0	v1.1
SCH-001	Event structure validation	Critical	Critical
UID-001	UUID v7 format	Critical	Critical
HCH-001	Genesis event prev_hash	Critical	Removed
HCH-003	Hash calculation algorithm	Critical	Critical (EventHash only)
SIG-001	Signature algorithm compliance	Critical	Critical
MKL-001	Merkle tree construction	-	Critical (New)
MKL-002	Merkle proof verification	-	Critical (New)
ANC-001	External anchor presence	-	Critical (New)
POL-001	Policy Identification	-	Critical (New)

9.1.3 Non-Critical Tests (NEW in v1.1)

Non-critical tests do not cause automatic certification failure but are reported in the certification report. Repeated non-critical failures may affect certification renewal.

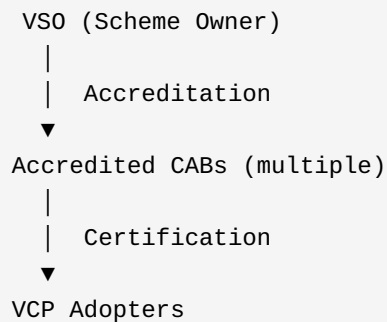
Test ID	Description	Silver	Gold	Platinum	Notes
HCH-002	Hash chain enabled (PrevHash)	Optional	Recommended	Recommended	For RTS25/ CAT alignment
ANC-002	Anchoring delay threshold	Warning >24h	Warning >1h	Warning >10min	Violation at 2x threshold
ANC-003	Anchor target availability	Check	Check	Check	Backup anchor recommended
CLK-001	Clock sync status consistency	Report	Verify NTP	Verify PTP	See Section 8.1

Test ID	Description	Silver	Gold	Platinum	Notes
XREF-001	Cross-reference ID uniqueness	Optional	Optional	Optional	If VCP-XREF enabled
XREF-002	Cross-reference reconciliation	Optional	Optional	Optional	If VCP-XREF enabled

NOTE on HCH-002: While PrevHash is OPTIONAL in v1.1, implementations targeting regulatory use cases (MiFID II RTS 25, SEC CAT Rule 613) SHOULD enable hash chain linking. Gold and Platinum tier implementations are RECOMMENDED to enable this feature.

9.2 Certification Governance

VC-Certified certification is issued by VSO-accredited Conformity Assessment Bodies (CABs), not by VSO directly. VSO functions as the **Scheme Owner**, responsible for standard development and CAB accreditation.



Entity	Role	Responsibilities
VSO	Scheme Owner	Standard development, CAB accreditation, test criteria
Accredited CAB	Certification Body	Certification issuance, conformance assessment

For detailed governance structure, see: **VSO-GOV-SCHEME-001** (VC-Certified Scheme Governance Structure)

10. Migration from v1.0

10.1 Backward Compatibility

VCP v1.1 is fully backward compatible with v1.0:

v1.0 Feature	v1.1 Behavior
Events with PrevHash	Fully supported, continues to work
Events without External Anchor	Must add anchoring (grace period available)
Events without Policy ID	Must add Policy ID (grace period available)

10.2 Migration Steps

For Silver Tier Implementations

- Add External Anchoring** (REQUIRED)
 - Implement daily Merkle root anchoring
 - Choose anchor target (OpenTimestamps recommended for simplicity)
 - Update Security object with MerkleRoot and AnchorReference
- Add Policy Identification** (REQUIRED)
 - Define PolicyID for your implementation
 - Add PolicyIdentification to all events
- Optional: Remove Hash Chain**
 - If hash chain adds complexity, it can now be removed
 - Set PrevHash to null or omit entirely

For Gold/Platinum Tier Implementations

- Update External Anchoring**
 - Ensure anchoring meets frequency requirements
 - Add AnchorReference to Security object
- Add Policy Identification** (REQUIRED)
 - Define PolicyID for your implementation
 - Add PolicyIdentification to all events
- Update Conformance Tests**
 - Remove HCH-001 (Genesis prev_hash) from critical tests
 - Add MKL-001, MKL-002, ANC-001, POL-001 to test suite

10.3 Grace Period

Implementations have the following grace period to comply with v1.1:

Requirement	Grace Period	Hard Deadline
External Anchor (Silver)	6 months	2026-06-25
Policy Identification	3 months	2026-03-25
Merkle fields in Security	3 months	2026-03-25

After the hard deadline, VCP v1.0-only implementations will not receive VC-Certified status.

11. Appendices

[Appendices largely unchanged from v1.0]

Appendix D: Three-Layer Architecture Summary (NEW)



Appendix E: Post-Quantum Cryptography Migration Guidance (Non-Normative)

This appendix provides non-binding guidance for implementations planning post-quantum cryptography (PQC) migration. These recommendations are informational and do not constitute v1.1 requirements.

E.1 Dual Signature Strategy

During the PQC transition period, implementations MAY use dual signatures to maintain backward compatibility while adding quantum resistance:

```
{
  "Security": {
    "Signature": "base64(Ed25519_signature)",
    "SignAlgo": "ED25519",
    "PQCSignature": "base64(Dilithium2_signature)",
    "PQCSignAlgo": "DILITHIUM2"
  }
}
```

E.2 Recommended Algorithm Combinations

Use Case	Classical	Post-Quantum	Notes
Standard	ED25519	DILITHIUM2	Balanced security/performance
Compact	ED25519	FALCON512	Smaller signatures
High Assurance	ED25519	DILITHIUM3	NIST Level 3

E.3 AnchorRecord PQC Extension

For external anchoring with PQC, extend the AnchorRecord schema:

```
{
  "AnchorRecord": {
    "MerkleRoot": "string",
    "Signature": "string",
    "SignAlgo": "ED25519",
    "PQCSignature": "string",
    "PQCSignAlgo": "DILITHIUM2",
    "Timestamp": "int64",
    "AnchorTarget": { ... }
  }
}
```

E.4 Migration Timeline Recommendation

Phase	Timeline	Action
Preparation	2025-2026	Implement dual-signature capability
Hybrid	2027-2029	Deploy dual signatures in production
Transition	2030+	Phase out classical-only signatures

NOTE: This timeline is advisory. Actual migration should be aligned with NIST PQC standardization progress and regulatory guidance.

12. References

Standards

- **RFC 9562:** Universally Unique IDentifier (UUID) v7
- **RFC 8785:** JSON Canonicalization Scheme (JCS)
- **RFC 6962:** Certificate Transparency
- **RFC 3161:** Time-Stamp Protocol (TSP)
- **IEEE 1588-2019:** Precision Time Protocol (PTP)
- **ISO 20022:** Universal financial industry message scheme

Regulations

- **MiFID II:** Markets in Financial Instruments Directive
- **RTS 24/25:** Regulatory Technical Standards
- **CAT Rule 613:** Consolidated Audit Trail
- **GDPR:** General Data Protection Regulation
- **EU AI Act:** Artificial Intelligence Act (2024)

Cryptography

- **FIPS 186-5:** Digital Signature Standard
- **FIPS 204:** Module-Lattice-Based Digital Signature Standard (Dilithium)
- **NIST SP 800-208:** Post-Quantum Cryptography
- **RFC 8032:** Edwards-Curve Digital Signature Algorithm (EdDSA)

Implementation

- **FIX Protocol:** Financial Information eXchange
- **SBE:** Simple Binary Encoding
- **FlatBuffers:** Memory Efficient Serialization Library
- **Apache Kafka:** Distributed Event Streaming

- **Redis Streams:** In-memory data structure store
-

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End of VeritasChain Protocol (VCP) Specification v1.1 DRAFT